

TED (15)-3151

Reg.No.....

(REVISION-2015)

Signature.....

THIRD SEMESTER DIPLOMA MODEL EXAMINATION IN
ENGINEERING/TECHNOLOGY-OCTOBER 2016

DATA COMMUNICATION

(Maximum marks: 100)

[Time: 3 hours

PART A

(Maximum marks: 10)

- I. Answer the following questions in one or two sentences. Each question carries 2 marks.
1. What is topology of a network?
 2. List different types of noises.
 3. Write two applications of coaxial cables.
 4. Define bit stuffing.
 5. Write any two advantages of full duplex communication. (5x2=10)

PART B

(Maximum marks: 30)

- II. Answer any five of the following questions. Each question carries 6 marks.
1. What are the components of communication? Explain.
 2. Compare baseband and broadband transmission.
 3. Write short note on Amplitude Shift Keying (ASK).
 4. List the advantages and disadvantages of optical fibre.
 5. Give a brief description about datagram circuit.
 6. Compare Character oriented and bit oriented protocols.
 7. Explain CSMA. (5x6=30)

PART C

(Maximum marks: 60)

(Answer one full question from each unit. Each full question carries 15 marks.)

UNIT--I

- III. a) Explain ISO-OSI model with a neat diagram. (15)

OR

- IV. (a) Explain different data flow methods. (7)
(b) What are the different network attributes? Explain. (8)

UNIT-II

- V. (a) What is multiplexing? Explain FDM. (8)
(b) Compare Synchronous and Asynchronous transmission. (7)

OR

- VI. (a) Write short note on
(i) WDM (ii) PSK (iii) bandwidth. (9)
(b) Compare AM, FM, PM. (6)

UNIT—III

- VII. (a) With a neat sketch, explain the constructional details and applications of twisted pair cable. (10)
(b) Write short note on cross bar switch (5)

OR

- VIII. (a) Briefly describe about circuit switching. (8)
(b) Explain the propagation modes of Fibre –optic cable. (7)

UNIT-IV

- IX. (a) What is a point to point protocol? Draw the PPP frame format and explain each field. (8)

(b) How parity-check code performs error detection? (7)

OR

X. (a) Illustrate cyclic redundancy Code with an example. (10)

(b) Explain stop and wait protocol. (5)